

Nikita Tatwadhia

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EDUCATION

University of Maryland, College Park, MD

Bachelor of Science in Bioengineering, *graduated with Honors in Engineering*

Completed May 2020

GPA: 3.6

SKILLS

Laboratory: Depth filtration, HPLC, SEC, Affinity chromatography, Biob⁺ testing, UF/DF, TFF, Protein assays, PCR, SDS-PAGE, Microscopy, BSC, Plating, Streaking, DNA extraction, TLC, Spectrophotometry, Analytical data entry, GDP, cGMP knowledge

Relevant Coursework: Biotechnology and Bioproduction, Biomedical Electronics, Fluids, Imaging, Statistics, Materials, Thermodynamics, Modeling Physiological Systems, Cell Biology and Physiology, Organic Chemistry, Statics, Calculus, Differential Equations

Software: MATLAB, SolidWorks, Autodesk Fusion 360, PTC Creo, Unicorn, Arduino, Fritzing, Microsoft Suite

Programming Languages: Python, HTML5, CSS, JavaScript

Languages: Punjabi (fluent), Hindi (proficient), Spanish (conservational)

WORK EXPERIENCE

Purification Process Sciences Intern

June – Aug. 2019

AstraZeneca, Gaithersburg, MD

- Utilized depth filtration, column chromatography, and low pH neutralization to recover mAbs in harvested cell culture
- Collected fractions of conditioned media that ran through three different depth filters to determine which had the biggest impact on reducing turbidity, DNA, and host cell protein concentrations
- Packed Omnifit columns and ran a scouting method on Unicorn to remove aggregates from multiple samples on an AKTA Pure
- Analyzed data to present findings at intern fair and department meetings

Undergraduate Researcher

May 2018 – May 2019

Bioinspired Advanced Manufacturing Lab, UMD-College Park, MD

- Worked under a graduate student in a team of 4 engineering students to design a microfluidics device capable of nanoparticle separation based off size via deterministic lateral displacement within the body
- Used 3D-printed microfluidics to apply liquid phase atomic layer deposition of thin films on nanoparticles
- Conducted extensive research to alter the design to utilize triangular nano posts and change the material of the device

ACADEMIC EXPERIENCE

Bioburden Detection on Surgical Instruments, Bioengineering Capstone Design

Aug. 2019 – May 2020

- Collaborated with 4 bioengineering students, 2 clinical mentors, and a faculty mentor to develop a method to easily detect biological residue on sterilized surgical instruments
- Designed a chemical soak consisting of fluorescamine and acetone to coat multiple tools, a viewing box to allow manipulation of various sized instruments, and an imaging software capable of detecting the presence and location of bioburden
- Individually tested different concentrations of fluorescamine on contaminated instruments and assembled final prototype of viewing box in Creo and Fusion 360
- Wrote multiple design reports and presented design to students and faculty in-person and virtually

Startup Venture and Business Plan, Entrepreneurship in Chemical and Life Sciences

Jan. – May 2019

- Worked with 4 bioengineering students to design a business model of a Class I medical device equipped with a gait monitoring system to track knee extension and flexion and provide feedback to physical therapists and patients with knee injuries
- Individually determined pricing, market analysis, funding plan, and promotional strategy
- Wrote a 15-page business plan and presented VC pitch to faculty, students, and potential investors

Over Sand Vehicle Project, Introduction to Engineering Design

Jan. – May 2017

- Collaborated in a team of 8 engineering students to program an OSV to navigate within 250 mm of debris, lift debris off sand, weigh debris within 20 g, and transmit correct type of material
- Individually designed circuitry and developed Arduino code to identify specific debris based on weight
- Successfully acquired, weighed, and communicated correct material of debris
- Wrote a 25-page formal design report and presented multiple milestones to faculty

AFFILIATIONS

Biomedical Engineering Society (BMES), *Active Member*

Aug. 2016 – May 2020

Society of Women Engineers (SWE), *Active Member*

Aug. 2016 – May 2020

Engineering Alumni Cup, *Bioengineering Team, Rube Goldberg Machine Competition*

Feb. 2019